



## Effects of facade appurtenances on the local pressure of high-rise building

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## ABSTRACT

The effects of façade appurtenances of high-rise building on local peak wind pressures were investigated through a series of wind tunnel experiments. Totally 21 appurtenances configurations simulated by thin horizontal splitter plates were studied. The unfavorable wind directions causing the largest peak pressure and the locations of the largest peak pressure were checked. It is noted that the locations of largest positive peak pressures can be strongly affected by the changes of vertical distance of plates. The area of large positive peak pressures of the models with appurtenances can be greatly smaller than the model without appurtenances. It is also found that the largest negative peak pressure on the higher leading corner can be mostly reduced by 42% from the smooth surface condition. By further checking the first four orders of moment of the probability density distribution of the pressure at upper leading corner, it is found that the splitter plates can strongly affect the characteristics of the probability distribution. The results also show that the appurtenances can not only reduce the negative local peak pressure of the higher leading corner on side face, but also dramatically reduce the variation range of negative peak pressure on the whole side face.

## 1. Introduction

With the high-rise buildings becoming slender and more flexible with the increase of height, they are more sensitive to wind load. Not only the overall effect including wind-induced dynamic vibration and base restraint force, but also the local effect such as local peak wind pressure, which should be carefully concerned by structural wind engineering researchers.

The mechanism of vortex formation and development for square-section high-rise buildings which might lead to serious dynamic responses has been explained in detail by many studies, including computational fluid dynamics (CFD) simulation and wind tunnel experiments (Peterka et al., 1985; Okuda and Taniike, 1993; Tominaga et al., 2008). Based on these results, many investigators started to focus on the measures to improve the aerodynamic performance by altering or suppressing the generation of vortex by modifying the shape of building, such as corner cut, chamfer, taper, set-back, openings and helical (Kwok et al., 1988; Tanaka et al., 2012; Kim et al., 2014). Tamura et al. (2017) investigated the aerodynamic characteristics of super-tall building models with various unconventional configurations based on a series of comprehensive experimental studies. They

reported that most configurations can reduce aerodynamic force, but helical models and polygon models do better. It has been proved that those approaches could reduce the wind-induced response and aerodynamic force effectively.

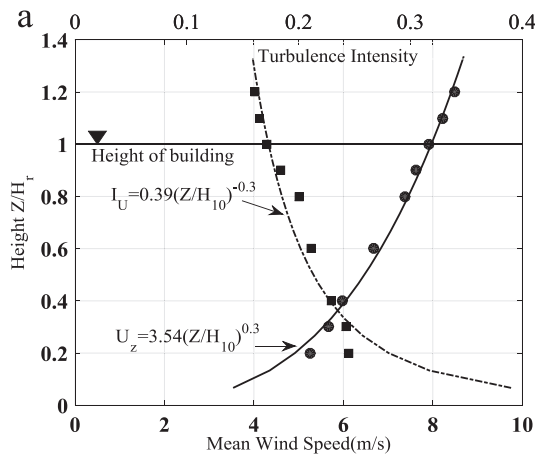
Some studies investigated the effect of local small scale changes of the building on local wind pressures. The earlier studies about local aerodynamic measure were made by Stathopoulos and Zhu (1988, 1990). They investigated the effect of uniform roughness, such as balconies and mullions, on the wind pressure. They reported that balconies with a width up to 4 m without walls reduce the pressure coefficient values on the building surface slightly. Conversely, mullions induce adverse wind effects at wall edges. They also made suggestions on the design of balcony based on their study. Maruta et al. (1998) reported effects of different balcony width on the local wind pressure and the balconies were fixed on each facade to measure the characteristics of the wind pressure variation in the same time. Reduction rate of peak pressure reached 0.67 when surface roughness magnitude equals to 1/10. The role of turbulence and the mechanism of local peak pressure reduction effect were also discussed in that study. More experiment investigations about surface roughness effect were made by many other researchers (Chand et al., 1998; Letchford et al., 2016).

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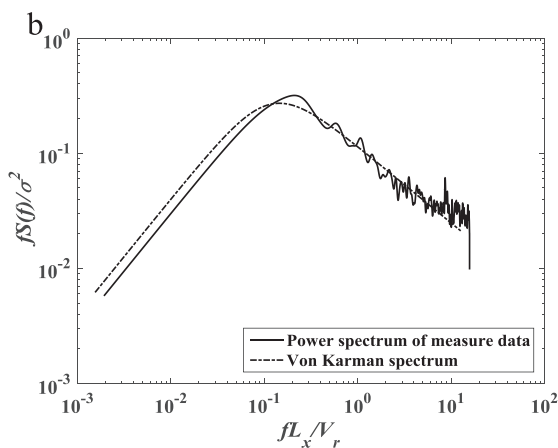
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Fig. 1. Wind tunnel setup and models.



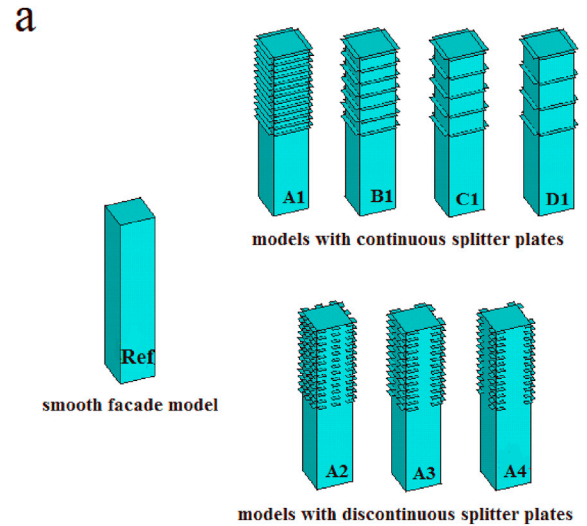
(a) Profile of mean wind speed and turbulence intensity



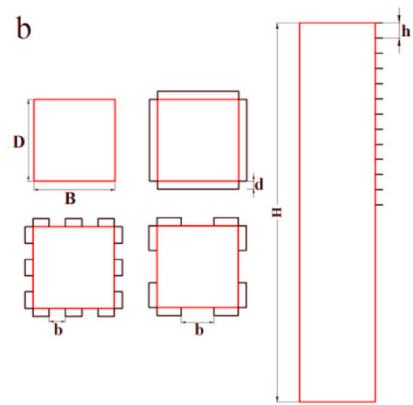
(b) Power spectrum of fluctuating wind speed

Fig. 2. Wind field information and characteristic in this study.

From the literature, surface roughness of building may reduce the local peak pressure on the side face by restraining the transmission of disturbances with approaching flow. However, only limited types of surface roughness were considered in previous study, and it can be



(a) Three-dimensional sketch of models



(b) Model configurations and appurtenances parameters

Fig. 3. Model information in this study.

expected that various arrangements of façade appurtenances may have quite different effects on wind pressure. In present study, more configurations of appurtenances are applied to investigate the effect of appurtenances on wind pressure systematically. Wind tunnel experiments were carried out based on a model of square section high-rise building and different appurtenances configurations are simulated by thin splitter plates horizontally attached to the facades. They are classified by horizontal gap distance ratios, vertical separation distance ratios and extensional depth ratios. Only the horizontal thin plates were adopted in this study, so that only the effects of appurtenances on the local wind load are studied, and the effects of change of cross-section shape are excluded. The relation between local extreme pressure coefficients and appurtenances configurations are investigated and discussed in details.

## 2. Wind tunnel experiments and model configurations

Wind tunnel experiments in this study were carried out in the boundary layer wind tunnel at Hunan University, China. The test section of the wind tunnel is 3 m wide, 2.5 m high and 11.5 m long. The urban terrain with power law index ( $\alpha$ ) of the mean wind speed profile approximately equaling to 0.3 (GB 50009-2012) is simulated through spires-and-roughness-blocks technique in the wind tunnel. The wind tunnel setup is shown in Fig. 1. The velocity scale is set to 1:8, and the geometric scale of 1:300 is adopted. As shown in Fig. 2(a), the mean wind

**Table 1**  
Parameters of the appendant.

| Category | $h$   | $h_r = h/H$ | $b$   | $b_r = b/D$ |
|----------|-------|-------------|-------|-------------|
| Ref      | /     | /           | /     | /           |
| A1       | 20 mm | 4%          | /     | /           |
| A2       | 20 mm | 4%          | 20 mm | 20%         |
| A3       | 20 mm | 4%          | 40 mm | 40%         |
| A4       | 20 mm | 4%          | 60 mm | 60%         |
| B1       | 40 mm | 8%          | /     | /           |
| C1       | 60 mm | 12%         | /     | /           |
| D1       | 80 mm | 16%         | /     | /           |

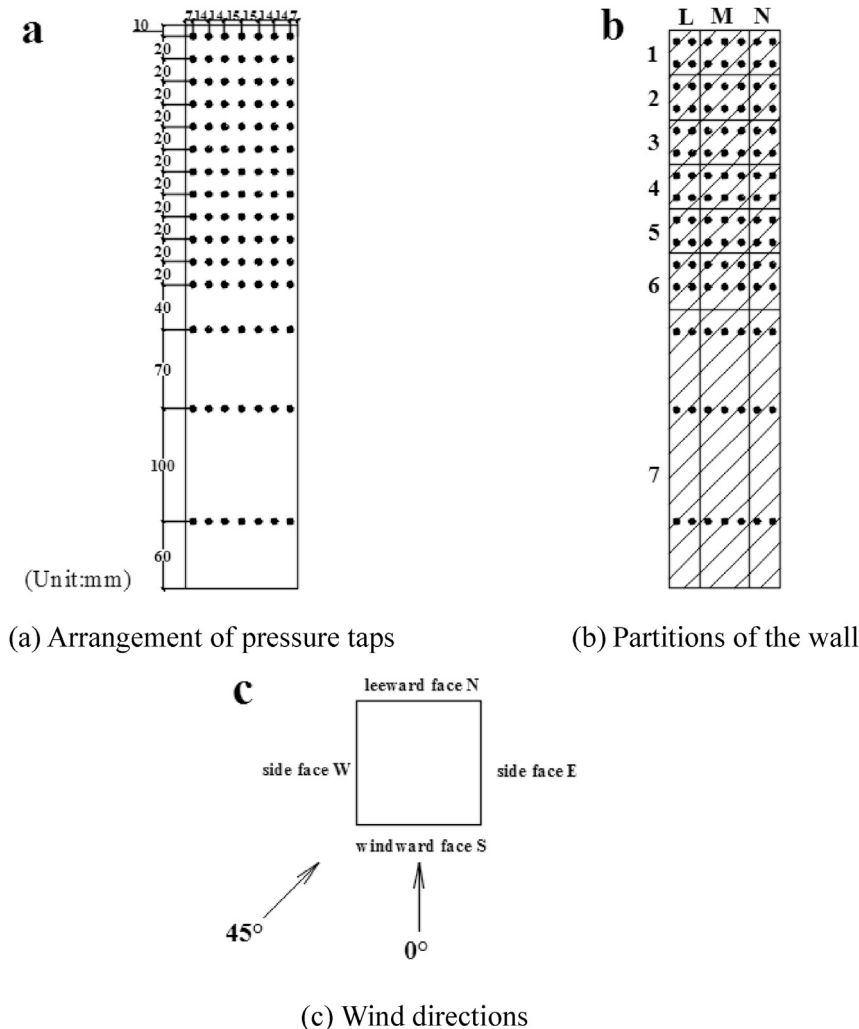
speed and turbulence intensity at the top height of the model are around 8 m/s and 17.2%, respectively. In this study, the turbulence integral length scale at the roof height of model is 0.41 m, which is an appropriate simulation of integral length scale in full scale (Hui et al., 2017). The Von Karman spectrum and the power spectrum of measured wind speed near roof of the model are plotted in Fig. 2(b), showing a good agreement.

8 types of rigid building model, including 1 reference model without appurtenances and 7 types of model with appurtenances as shown in Fig. 3(a), were tested in presented study. All of the models have the same square cross section with  $B \times D$  of 100 mm  $\times$  100 mm and the same height ( $H$ ) of 500 mm. Different horizontal thin plates were attached to the four sides of the building facade to simulate different appurtenances configurations. As illustrated in Fig. 3(b), the appurtenances are

classified based on three parameters, where “ $d$ ” indicates the extensional depth of the thin plates, “ $b$ ” means the horizontal gap between two adjacent plates and “ $h$ ” represents the vertical separation between two adjacent plates. The above parameters are expressed in the form of non-dimensional ratio, as shown in Table 1. Model Ref is the model without any appurtenance used as the reference model, model A1, B1, C1 and D1 have continuous plates with different vertical separations, and model A2, A3 and A4 have discontinuous plates with different horizontal gaps. In addition, there are three different values of  $d_r$  ( $d_r = d/D$ ) for each arrangement, which are 7.5%, 10% and 12.5%, respectively. Roman numerals are used to indicate different  $d_r$ , such as A1-I (7.5%), A1-II (10%) and A1-III (12.5%). Hence, a total of 22 sets of tests were conducted. Since wind load on the upper part of building is more critical for structure design, the appurtenances plates in this study were only equipped within the upper part of the building as shown in Fig. 3.

The arrangement of pressure taps on model surface and the definition of wind directions ( $\theta$ ) in the experiments are shown in Fig. 4. A total of 420 pressure taps were fixed at 15 levels on the wall, with each level 28 taps as shown in Fig. 4(a). In order to make a detailed checking of the effects of appurtenances, more taps were set at the upper part of the model. In addition, each wall is divided into 21 partitions as shown in Fig. 4(b) to clarify the position of peak pressure. Wind directions in this study for each type of model were from  $0^\circ$  to  $45^\circ$  with  $5^\circ$  interval due to the symmetry of the models and shown in Fig. 4(c).

In present study, a multi-channel simultaneous fluctuating pressure measurement system was used to obtain the time-histories of fluctuating



**Fig. 4.** Detailed experiment and model information.

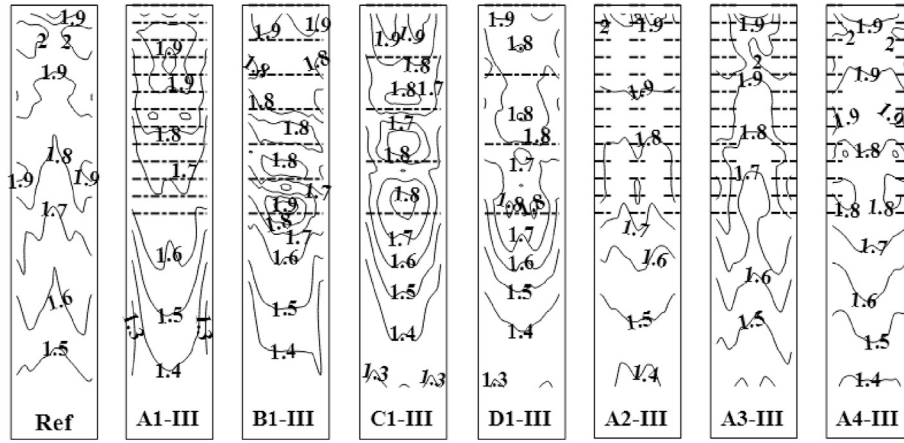


Fig. 5. The contours of  $C_{P,max}(i)$  on the windward face S ( $h_r = 12.5\%$ ).

pressure, with sampling frequency 312.5 Hz. Since time scale is 1:37.5, fifteen 16-s long samples, corresponding to fifteen 10-min long samples in full scale, were collected for each wind direction.

The tubing effect of the measurement system was compensated by using the method proposed by Irwin (Irwin et al., 1979). Moreover, moving average filter (Holmes, 1997) were employed to filter the fluctuating pressure in this study to convert the measured point pressure into the value representing the wind pressure acting on the corresponding area. The averaging time  $\tau$  is defined as the following formula:

$$\tau = 1.0 \times L/U_H \quad (1)$$

Where  $\tau$  is around 0.12 s in full scale based on the sampling rate,  $U_H = 64$  m/s is the design wind speed at building height in full scale and  $L$  is the length of the diagonal for the panel of interest.

The time history of wind pressure at each tap is expressed as a non-dimensional pressure coefficient  $C_P(i, j, \theta, t)$ , where  $i$  is the sequence number of the tap ( $i = 1, 2 \dots 420$ ),  $j$  indicates the sequence number of the studied sample ( $j = 1, 2 \dots 15$ ),  $\theta$  is the wind direction ( $\theta = 0, 5, 10 \dots 360$ ) and  $t$  is the time instance. The first four orders of moment of the pressure coefficient time histories  $C_{P\_XXX}(i, \theta)$  are then obtained from the ensemble average as following:

$$C_{P\_XXX}(i, \theta) = \frac{1}{15} \sum_{j=1}^{15} C_{P\_XXX}(i, j, \theta) \quad (2)$$

Where XXX can be “mean”, “std”, “skew”, or “kur” if mean, standard deviation, skewness, or kurtosis of the random processes need to be calculated.

For the reference model, the largest mean pressure coefficient of 0.90 is checked at wind direction of  $0^\circ$  and the smallest mean pressure coefficient is observed as  $-0.97$  with wind direction of  $10^\circ$ . The largest fluctuating pressure coefficient at side surfaces is 0.426 at wind direction

of  $0^\circ$ . Similar observations were reported by Kim and Kanda (2013).

The peak pressure coefficient at tap  $i$  in direction  $\theta$  is obtained by using the Cook and Mayne (1980) method as follows:

$$C_{P\_peak}(i, \theta) = U_{C_{P\_peak}}(i, \theta) + 1.4 / a_{C_{P\_peak}}(i, \theta) \quad (3)$$

Where *peak* becomes “max” for positive peak and “min” for negative peak,  $U_{C_{P\_peak}}(i, \theta)$  and  $a_{C_{P\_peak}}(i, \theta)$  are the location and scale parameter of the extreme pressure coefficient, which can be estimated based on Cook-Mayne method (Cook and Mayne, 1980). The constant 1.4 in Eq. (2) is the reduced variate corresponding 50 year return period design.

### 3. Experiment results and discussion

#### 3.1. Effects of appurtenances on the positive peak pressure

$C_{P,max}(i)$  is defined as the largest positive peak pressure coefficients of tap  $i$  in all wind directions, which can be calculated as following equation:

$$C_{P,max}(i) = \max_{0^\circ \leq \theta \leq 360^\circ} \{C_{P,max}(i, \theta)\} \quad (4)$$

The location of the largest positive peak pressure coefficients of the whole windward face can be obtained by calculating the maximum value of  $C_{P,max}$  as following:

$$C_{P,max} = \max_{1 \leq i \leq 420} \{C_{P,max}(i)\} \quad (5)$$

The contours of  $C_{P,max}(i)$  of the South wall of different models with  $h_r = 12.5\%$  are shown in Fig. 5, where the dashed lines stand for the appurtenances plates on surface. Comparison with the results of model without appurtenance shows that the appurtenances does not severely

**Table 2**  
The largest positive peak pressure coefficients.

| -I  |             |           |            | -II  |             |           | -III |             |            |
|-----|-------------|-----------|------------|------|-------------|-----------|------|-------------|------------|
|     | $C_{P,max}$ | Partition | Area ratio |      | $C_{P,max}$ | Partition |      | $C_{P,max}$ | Area ratio |
| A1  | 2.10        | L1/N1     | 48.3%      | 1.93 | M2          | 26.3%     | 1.96 | M1          | 20.7%      |
| B1  | 1.99        | M5        | 43.8%      | 1.90 | L1/N1       | 13.6%     | 1.94 | M6          | 30.2%      |
| C1  | 2.06        | M4        | 46.1%      | 1.96 | M2          | 29.7%     | 1.92 | M1          | 13.7%      |
| D1  | 1.97        | L1/N1     | 25.6%      | 1.94 | L1/N1       | 22.3%     | 1.92 | L1/N1       | 21.3%      |
| A2  | 2.03        | M1        | 34.1%      | 2.02 | L1/N1       | 38.1%     | 2.04 | L1/N1       | 35.4%      |
| A3  | 2.11        | L1/N1     | 39.8%      | 2.02 | L1/N1       | 42.1%     | 2.10 | L1/N1       | 36.9%      |
| A4  | 2.04        | L1/N1     | 38.1%      | 2.00 | L1/N1       | 31.9%     | 2.02 | L2/N2       | 37.0%      |
| Ref | 2.04        | M1        | 40.4%      |      |             |           |      |             |            |

Note: Area ratio means the ratio of the area with corresponding  $C_{P,max}(i) > 1.8$  to the whole area of a wall.

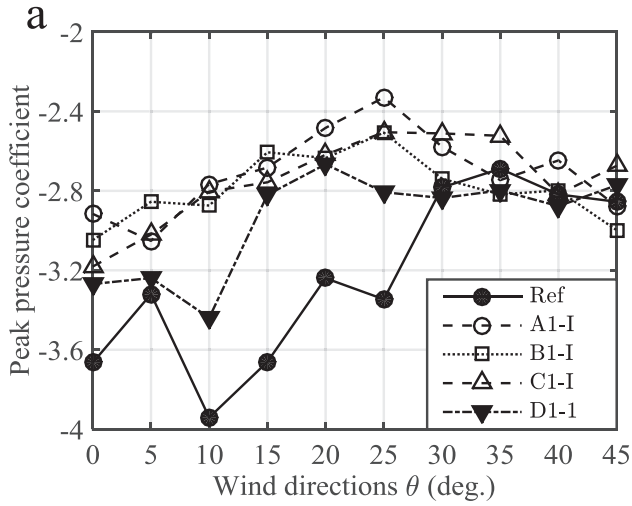
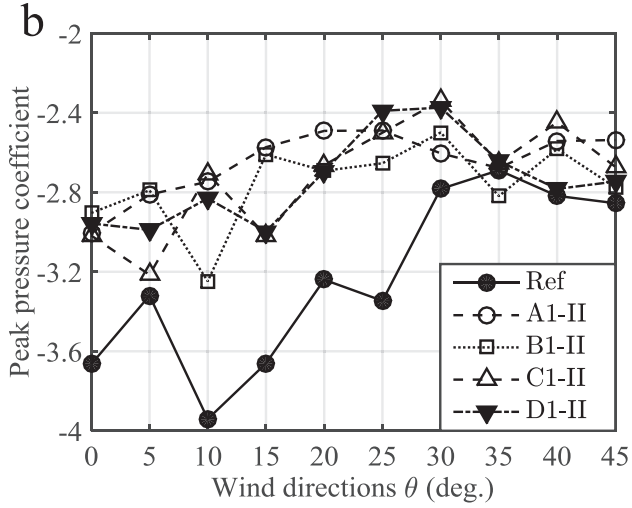
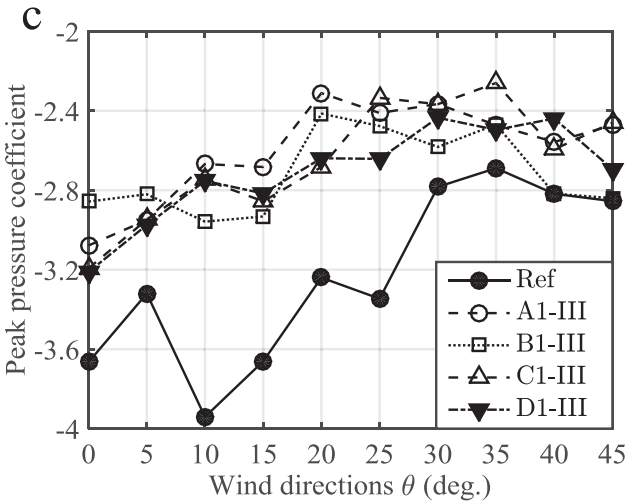
(a) Variation of  $C_{P\_min}(\theta)$  for  $d_r=7.5\%$ (b) Variation of  $C_{P\_min}(\theta)$  for  $d_r=10\%$ (c) Variation of  $C_{P\_min}(\theta)$  for  $d_r=12.5\%$ 

Fig. 6. Largest negative peak pressure coefficients for each wind direction (continuous splitter plates).

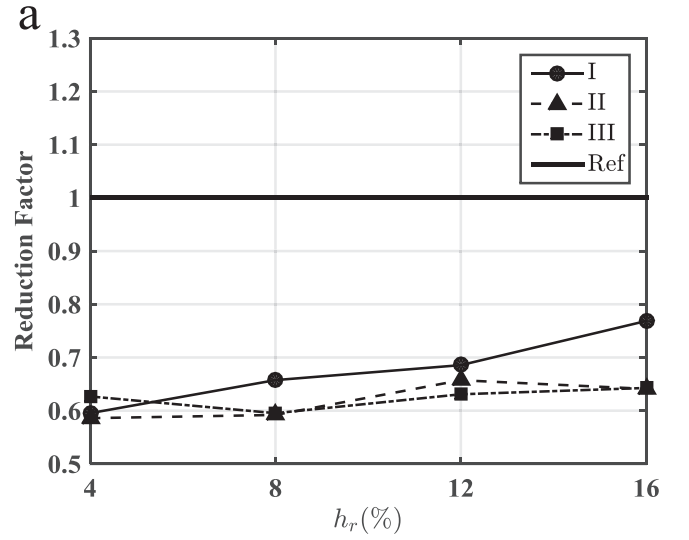
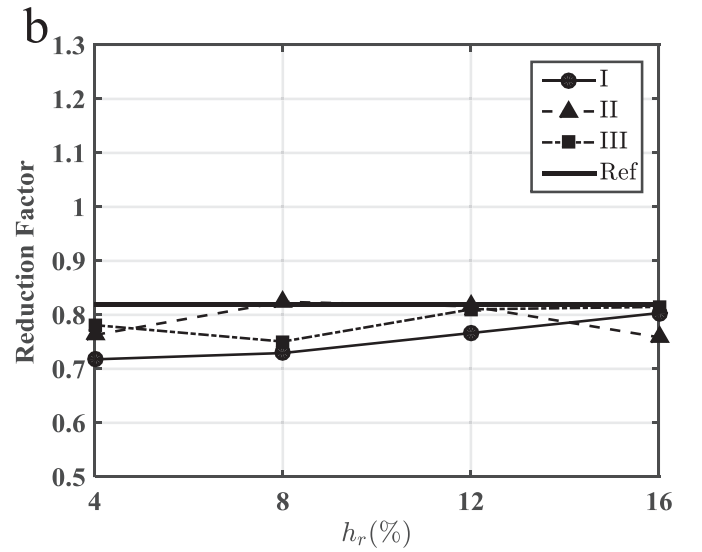
(a)  $RF_U$ (b)  $RF_L$ 

Fig. 7.  $RF$  for different vertical separation distances ratios.

affect the distribution pattern and values of  $C_{P\_max}(i)$  except for the model B1-III and C1-III. According to the results of the two models (B1-III and C1-III), the stagnations can be found in between almost every two adjacent plates. Such multi-stagnation results indicate that within an appropriate range of the vertical gaps ratios between two adjacent plates ( $h_r \in [8\%, 12\%]$ ), the flows in front of the windward wall are isolated in each cell, no vertical transmission between different cells are allowed, leading to the fact that stagnation happens within each cell. By checking the results of models with smaller  $d_r$ , it is found that such multi-stagnation phenomenon becomes weak with the decreasing of  $d_r$  value. Thus, it can be known that the larger depth ( $d_r$ ) of the appurtenances plates, the stronger the effect they may have on the flow pattern. At the same time, the multi-stagnation phenomenon is not observed on the models with discontinuous plates, indicating that such appurtenances arrangement does not have the flow isolation effect.

More details about the positive peak pressure coefficients of each model are shown in Table 2. It can be checked that the  $C_{P\_max}$  on the



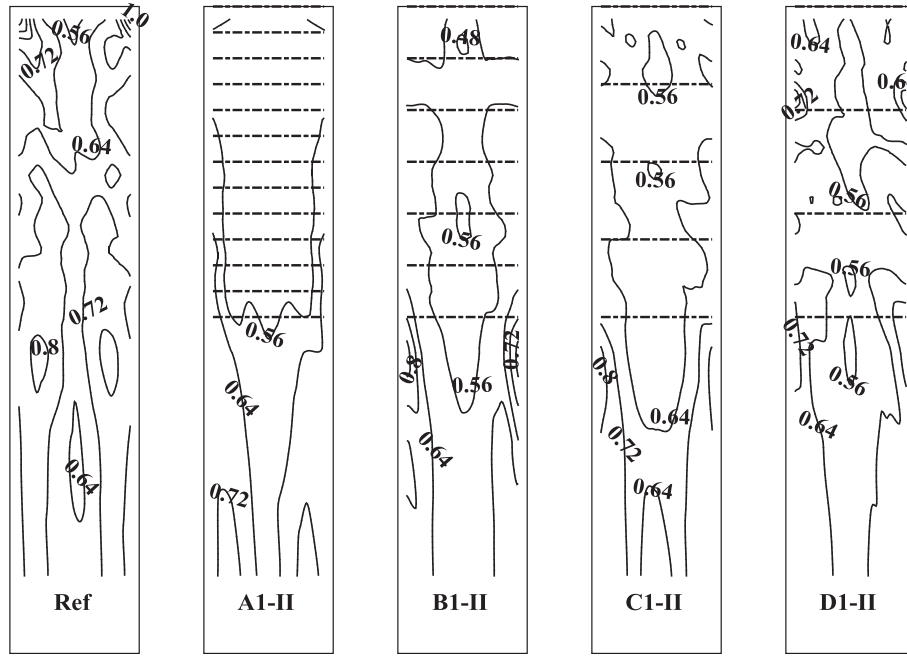


Fig. 8. The distributions of  $RF(i)$  for models in different  $h_r$  ( $d_r = 10\%$ ).

Table 3

The mean, standard deviation and maximum value of  $RF_U(i)$  (continuous plates).

|     | -I          |            |            | -II         |            |            | -III        |            |            |
|-----|-------------|------------|------------|-------------|------------|------------|-------------|------------|------------|
|     | $RF_{mean}$ | $RF_{std}$ | $RF_{max}$ | $RF_{mean}$ | $RF_{std}$ | $RF_{max}$ | $RF_{mean}$ | $RF_{std}$ | $RF_{max}$ |
| A1  | 0.58        | 0.059      | 0.76       | 0.54        | 0.045      | 0.69       | 0.56        | 0.045      | 0.69       |
| B1  | 0.59        | 0.035      | 0.67       | 0.56        | 0.034      | 0.68       | 0.58        | 0.031      | 0.67       |
| C1  | 0.63        | 0.046      | 0.76       | 0.61        | 0.041      | 0.70       | 0.60        | 0.038      | 0.68       |
| D1  | 0.63        | 0.061      | 0.87       | 0.60        | 0.050      | 0.76       | 0.62        | 0.045      | 0.72       |
| Ref | 0.68        | 0.077      | 1          |             |            |            |             |            |            |

whole face varies from 1.90 (B1-II) to 2.11 (A3-I), while this value is 2.04 for the model Ref. Such results indicate that the appurtenances do not have very strong effects on the value of  $C_{P,max}$ . At the same time, the lists of partition in Table 2 are based on Fig. 4(b) representing the locations of  $C_{P,max}$ . The results indicate that although the locations of  $C_{P,max}$  for different models are all observed at the upper part of the wall, the presence of the appurtenances can strongly affect the location of  $C_{P,max}$ , especially for model B1 and C1 because of the multi-stagnation phenomenon. The ratio of the area with corresponding  $C_{P,max}(i)$  greater than 1.8 to the whole area of a wall is also studied. It is found that the ratios of most of the models with appurtenances are lower than the reference model whose value is about 40%. Such results mean that when the building surface is equipped by the appurtenances, the area of cladding with high design positive peak pressure can be reduced. When  $d_r = 7.5\%$ , the ratio of model D1 is only about 25%. With the greater  $d_r$  values, the ratios of model B1-II and C1-III can even be lower than 14%, indicating a reduction of design wind load value for quite a large area. Comparison with the models with continuous splitter plates, results show the discontinuous splitter plates (A2, A3 and A4) have relatively weak effects on the positive peak pressures.

### 3.2. Effects of appurtenances on the negative peak pressure

For a square-section high rise building, the large negative local peak pressures are always found on the side face. Therefore, it is necessary to have a detailed study on the negative peak pressure for the cladding design.

#### 3.2.1. Effect of vertical separation distance ( $h_r$ )

To study the effect of vertical separation, one largest negative peak pressure coefficient  $C_{P,min}(\theta)$  is chosen from all of points in each wind direction of each model equipped with continuous splitter plates, as shown in Fig. 6.  $C_{P,min}(\theta)$  and  $C_{P,min}(i)$  can be calculated as follows:

$$C_{P,min}(\theta) = \min_{1 \leq i \leq 420} \{C_{P,min}(i, \theta)\} \quad (6)$$

$$C_{P,min}(i) = \min_{0^\circ \leq \theta \leq 360^\circ} \{C_{P,min}(i, \theta)\} \quad (7)$$

To quantify the appurtenances effect, one largest value  $C_{P,min}$  is selected from all points and all directions on the side face, which can be obtained as follows:

$$C_{P,min} = \min_{0^\circ \leq \theta \leq 360^\circ} \{C_{P,min}(\theta)\} \quad (8)$$

The results of the model Ref are also shown in Fig. 6, due to symmetry property, only the results from  $0^\circ$  to  $45^\circ$  are shown. One can see that although  $C_{P,min}$  always occurs in the range of  $0^\circ$ – $15^\circ$  for both reference and studied models, the  $C_{P,min}$  of the models with appurtenances occurs at the wind direction more close to  $0^\circ$ , and for many cases  $\theta = 0^\circ$ . At the same time, it can also be checked that the  $C_{P,min}$  of model Ref is  $-3.94$ , while the values of the models with appurtenances are much lower than this value and mostly vary between  $-2.8$  and  $-3.2$ . The severe decrement of  $C_{P,min}$ , which is more than 20%, suggests the significant effects of the appurtenances on the negative peak pressures.

The locations  $C_{P,min}$  occurs are also concerned in this study. For the reference model, relative large negative peak values can be observed at

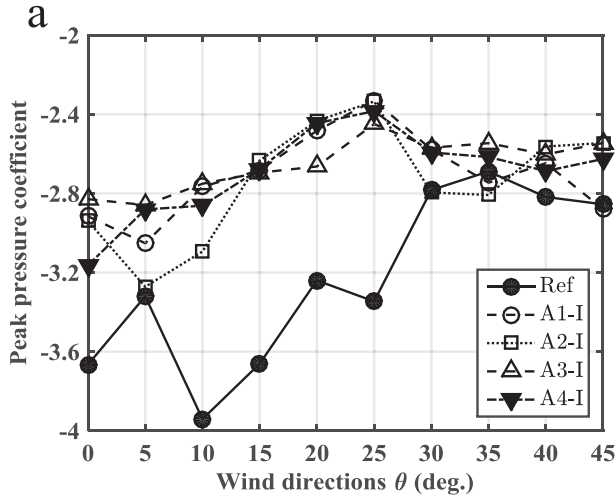
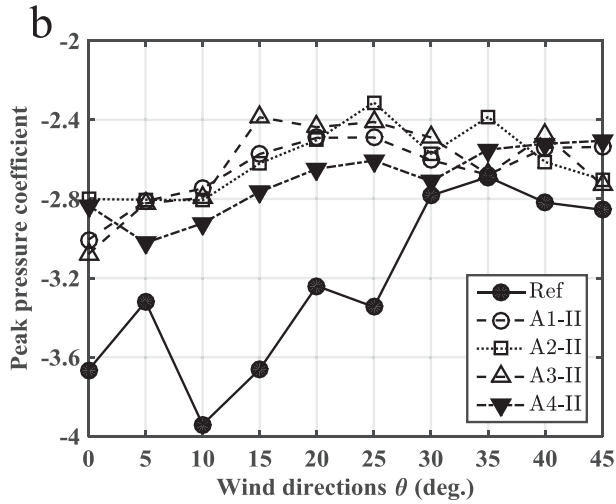
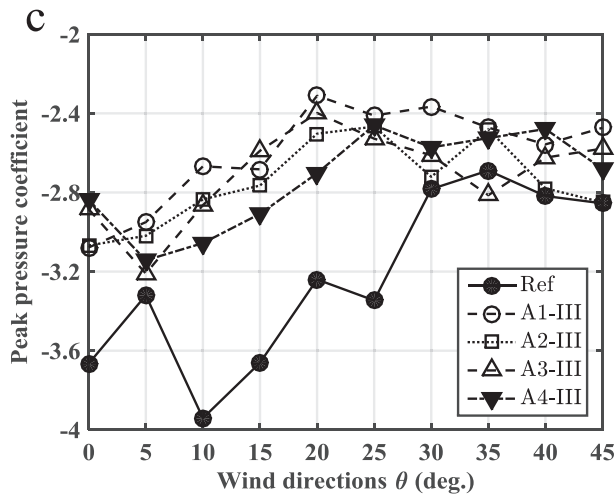
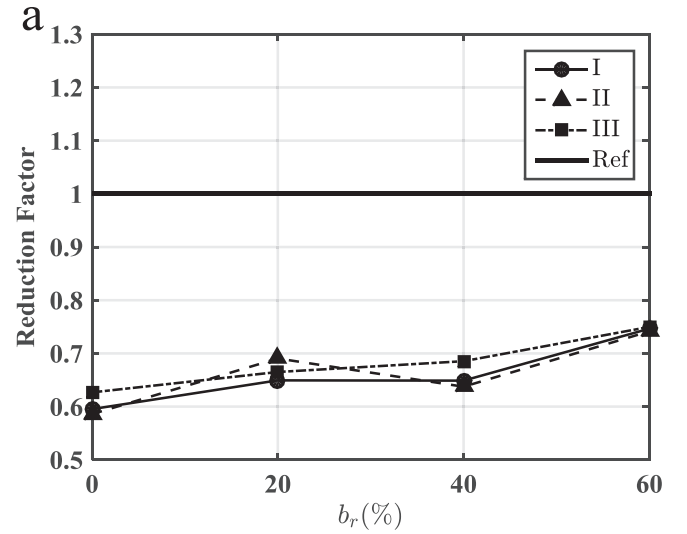
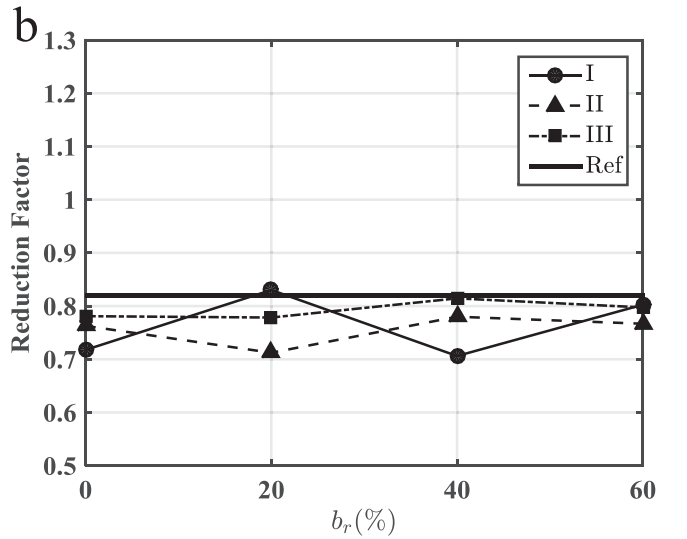
(a) Variation of  $C_{P\_min}(\theta)$  for  $d_r=7.5\%$ (b) Variation of  $C_{P\_min}(\theta)$  for  $d_r=10\%$ (c) Variation of  $C_{P\_min}(\theta)$  for  $d_r=12.5\%$ 

Fig. 9. Largest negative peak pressure coefficients for each wind direction (discontinuous plates).

(a)  $RF_U$ (b)  $RF_L$ Fig. 10.  $RF$  for different horizontal separation distances ratios.

upper leading corner and lower leading edge. The similar phenomenon was also observed by Kim and Kanda (2013). In present study,  $C_{P\_min}$  of reference model ( $-3.94$ ) is observed at the upper leading corner, which is expressed as partition N1 (L1) at Fig. 4 (b). Due to the effects of appurtenances attached at the upper part of the building, the negative peak pressure at upper leading corner on the models with appurtenances is significantly reduced. However, the largest negative peak values at lower leading edge are not severely affected. The reduction factor ( $RF$ ) at upper leading corner (N1) or lower leading edge (N7) is adopted to indicate the appurtenances effect on negative peak pressure coefficients as follow:

$$RF_X = \frac{C_{P\_min\_X}}{C_{P\_min\_Ref}} \quad (9)$$

Where  $C_{P\_min\_X}$  indicates that the largest negative peak pressure coefficients at upper leading corner or lower leading edge.  $X$  becomes “U” for upper leading corner and “L” for lower leading edge,  $C_{P\_min\_Ref}$

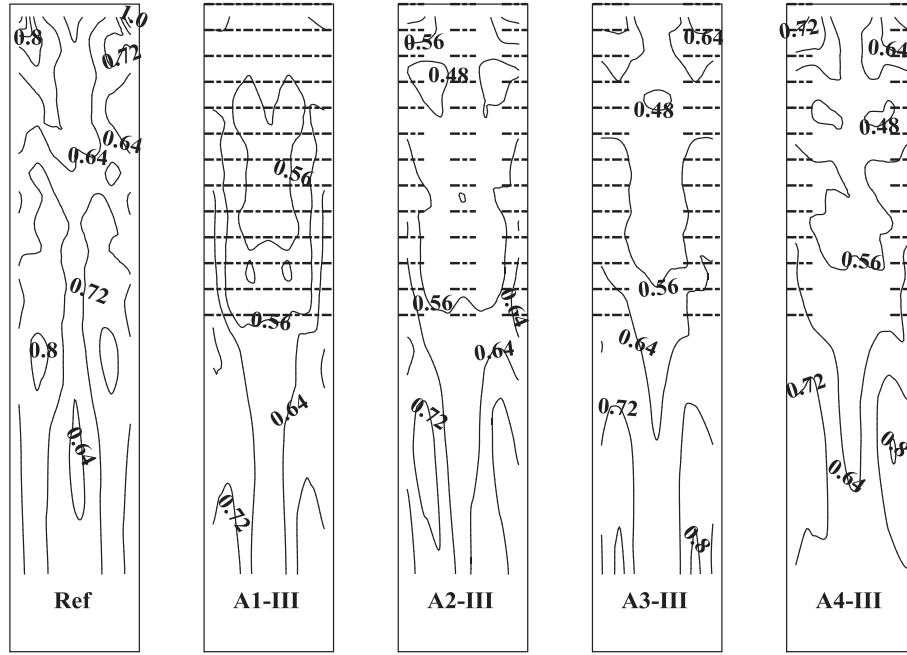


Fig. 11. The distributions of  $RF(i)$  for models in different  $b_r$  ( $d_r = 12.5\%$ ).

Table 4

The average value, standard deviation and maximum value of  $RF_U(i)$  (discontinuous plates).

|     | -I          |            |            | -II         |            |            | -III        |            |            |
|-----|-------------|------------|------------|-------------|------------|------------|-------------|------------|------------|
|     | $RF_{mean}$ | $RF_{std}$ | $RF_{max}$ | $RF_{mean}$ | $RF_{std}$ | $RF_{max}$ | $RF_{mean}$ | $RF_{std}$ | $RF_{max}$ |
| A1  | 0.58        | 0.059      | 0.76       | 0.54        | 0.045      | 0.69       | 0.56        | 0.045      | 0.69       |
| A2  | 0.56        | 0.054      | 0.73       | 0.55        | 0.055      | 0.69       | 0.55        | 0.053      | 0.70       |
| A3  | 0.55        | 0.049      | 0.68       | 0.55        | 0.053      | 0.67       | 0.56        | 0.055      | 0.69       |
| A4  | 0.59        | 0.066      | 0.75       | 0.57        | 0.058      | 0.74       | 0.57        | 0.062      | 0.75       |
| Ref | 0.68        | 0.077      | 1          |             |            |            |             |            |            |

indicates the largest negative peak pressure coefficients of reference model which is  $-3.94$  in this study.

The  $RF_U$  and  $RF_L$  for different kinds of models are shown in Fig. 7. With the comparison of the results from model Ref in Fig. 7(a), one can clearly see that when the building is equipped by the appurtenances, the largest negative peak pressure at the upper leading corner of the side face can be drastically decreased with  $RF_U \in [0.6, 0.8]$ . The minimum  $RF_U$  (0.586) was observed for model A1-II ( $d_r = 10\%$ ,  $h_r = 4\%$ ). Such results mean that  $C_{p,min,U}$  can be reduced by 20%–40%, indicating the strong effects of the appurtenances on the negative peak pressures. It can also be noted that when  $d_r = 7.5\%$ , the variation of  $RF_U$  with the increasing of  $h_r$  tends to be greater than the cases  $d_r = 10\%$  and  $12.5\%$ . For the models with cases  $d_r = 10\%$  and  $12.5\%$ , when  $h_r$  varies from 4% to 16%,  $RF_U$  remain in between 0.6 and 0.7. In other words, the vertical separation distance of the appurtenances has little effect on the largest negative peak pressure at upper leading corner. At the same time, from Fig. 7(b) one can see that  $RF_L$  do not show big differences between the model Ref and the models with appurtenances. The minimum  $RF_L$  (0.72) is observed for model A1-I ( $d_r = 7.5\%$ ,  $h_r = 4\%$ ) and  $RF_L$  of the model Ref is 0.81. Such results indicate that the appurtenances equipped at the upper part do not have strong influence on the negative peak pressure at the lower leading edge of the building.

Apart from largest negative peak pressure, effects of appurtenances on the distribution of negative peak pressure of whole side face are also studied. Models equipped with different types of continuous plates are considered and reduction factor of each tap  $RF(i)$  is calculated through the following equation:

$$RF(i) = \frac{C_{p,min}(i)}{C_{p,min,Ref}} \quad (10)$$

Fig. 8 shows the distributions of  $RF(i)$  on side face of models with different vertical separation distances of appurtenances when  $d_r = 10\%$ . It is remarkable that the negative peak pressure on reference model varied severely with  $RF(i)$  varying from 0.56 to 1.0. For models with appurtenances,  $RF_U(i)$  (values of  $RF(i)$  at the upper part) are much smaller than those values of model Ref. At the same time, the variation of  $RF_U(i)$  becomes more and more inconspicuous with the decrease of  $h_r$ , especially for model A1-II ( $0.56 < RF_U(i) < 0.64$ ) and B1-II ( $0.48 < RF_U(i) < 0.64$ ). However, for the lower part without appurtenances, the values of  $RF_L(i)$  are slightly affected and vary mostly between 0.56 and 0.8 for different vertical separations, which are the same as the model Ref.

As discussed above, the negative peak pressure on lower part of the models with appurtenances are similar to the reference model due to the absence of appurtenances. Thus, to investigate the appurtenances effect more accurately, all the pressure taps on the side face except for partition L7, M7, and N7 are selected to calculate the mean, standard deviation and maximum value of  $RF_U(i)$ .  $RF_{mean}$ ,  $RF_{std}$  and  $RF_{max}$  are applied to represent those values respectively.

As shown in Table 3, the  $RF_{mean}$  of model Ref is 0.68, while those values of models with appurtenances are much smaller than it. Especially for the model A1-II,  $RF_{mean}$  even decreases to 0.54. Moreover, it is found that the  $RF_{mean}$  slightly increase with the increase of  $d_r$ . These results indicate that the negative peak pressure of most taps at the upper part of wall can be significantly reduced due to the appurtenances. At the same



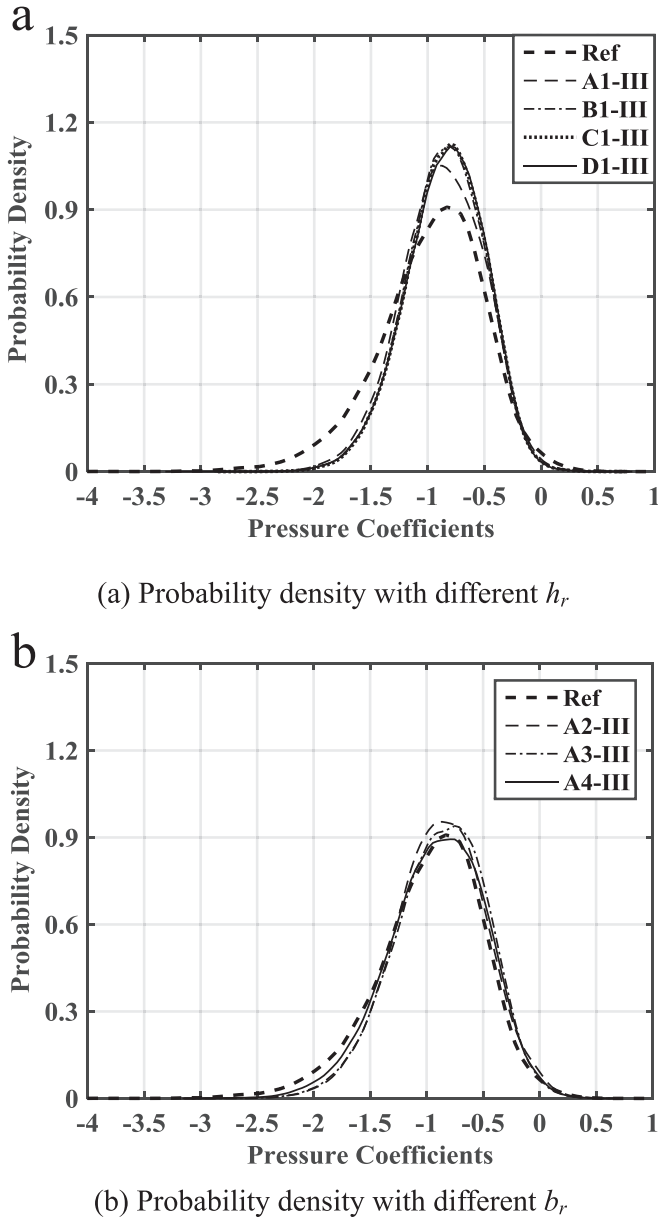


Fig. 12. The probability density of fluctuating pressure at the leading corner.

time, the effect of appurtenances on  $RF_{std}$  is also clearly observed from Table 3. The minimum  $RF_{std}$  of model with appurtenances (0.031) is barely half of the  $RF_{std}$  of model Ref (0.077) clearly indicating a low variation of negative peak pressures on the wall. One can also find that the value of  $RF_{std}$  becomes smaller when  $d_r$  increases, suggesting that the effect of appurtenances on restraining the variations of negative peak pressure is stronger when the depth of splitter plates is larger. The differences of  $RF_{max}$  of different models with appurtenances are not obvious, especially when  $h_r$  increase from 4% to 12%, and  $RF_{max}$  basically maintains at 0.7. It can be easily checked that,  $RF_{max}$  of the models with appurtenances are much smaller than that of model Ref with maximum decrement of 33%.

### 3.2.2. Effect of horizontal separation distance ( $b_r$ )

The experiment results of the models with constant  $h_r$  (4%) and four different  $b_r$  (0% (A1), 20% (A2), 40% (A3), and 60% (A4)) are adopted. The largest negative peak pressure coefficients  $C_{p,min}(\theta)$  and  $C_{p,min}$  are used again to quantitatively investigate the effects of horizontal gaps on

negative peak pressure as shown in Fig. 9. One can see that the most unfavorable wind direction  $\theta$  ( $\theta$  that causes the largest  $C_{p,min}$ ) of model equipped by discontinuous splitter plates are mostly lower than  $5^\circ$ , and the  $C_{p,min}$  of models with appurtenances reduce severely from that value of model Ref. However, comparing with type A1 models, models equipped by discontinuous appurtenances always have relative lower values of  $C_{p,min}$ .

Reduction factor (RF) at the two chosen partitions N1 and N7 of models with different  $b_r$  are shown in Fig. 10, where the results of model Ref are also plotted for reference. In Fig. 10 (a), one can clearly see that the values of  $RF_U$  are much smaller than 1 and vary in the range of 0.6–0.75. Such results indicate that the effect of appurtenances on the negative peak pressures is strong even though the plates are not continuous. It can also be checked that the values of  $RF_U$  become relatively larger when  $b_r$  increases from 0% to 60%, suggesting that smaller horizontal gaps lead to more significant decrement effect on  $C_{p,min}$  at upper leading corner. Fig. 10 (b) shows that  $C_{p,min}$  at the lower leading edge of models with appurtenances vary barely with  $RF_L \in [0.7, 0.8]$  and close to that of model Ref ( $RF_L = 0.81$ ). It can be concluded that  $C_{p,min}$  at the lower leading edge are influenced slightly by the distance of horizontal gaps. Moreover, the variations of  $RF_L$  with the change of  $b_r$  do not show clear dependence on plate depth. When  $d_r = 7.5\%$ ,  $RF_L$  is small for type A1 and A3 but much larger for type A2 and A4. An inverse trend, however, is observed in case  $d_r = 10\%$  and  $d_r = 12.5\%$ .

The effect of horizontal gaps on  $C_{p,min}(i)$  of whole side face is also studied by adopting  $RF(i)$  again. Fig. 11 shows the distributions of  $RF(i)$  on side face of models with different horizontal in case  $d_r = 12.5\%$ . It can be checked  $RF_U(i)$  of the models with appurtenances are much smaller than those values of model Ref with  $RF_U(i) \in [0.56, 1.0]$ , which are similar as the results observed from models B1, C1, D1. The values of  $RF_U(i)$  are mainly kept around 0.56 when  $b_r = 0$ , but the  $RF_U(i)$  vary from 0.50 to 0.72 when  $b_r = 60\%$ . That means the variation of  $RF_U(i)$  becomes stronger with the increase of  $b_r$ . By further checking the lower part of the side face, it is found again that the models with appurtenances share a similar distribution pattern of  $RF_L(i)$  with model Ref.

Table 4 shows the mean, standard deviation and maxima of  $RF_U(i)$  for different models. It can be clearly checked that  $RF_{mean}$ ,  $RF_{std}$  and  $RF_{max}$  of models with appurtenances are all much smaller than those of model Ref. In Sec. 3.2.1, it is found that the variation of  $RF_U(i)$  is sensitive to the vertical separation of plates. But for models with same vertical separation distance,  $RF_{mean}$  and  $RF_{max}$  have similar values, which are 0.56 and 0.75, respectively when  $d_r = 7.5\%$ . The two values become 0.56 and 0.7, when  $d_r = 10\%$ ,  $d_r = 12.5\%$  as shown in Table 4. At the same time, the value of  $RF_{std}$  shows clear differences for different models. The largest value is always observed on A4, and the smallest value is obtained from A1. Such results indicate that the effect of discontinuous plates on restraining the variation of negative peak pressure is not as significant as continuous plates.

### 3.3. Probability density of wind pressure coefficients

The Non-Gaussianity of probability of pressure on square section buildings investigated by Ko et al. (2005) indicated that the Non-Gaussian wind pressure on the side face is caused by relatively large suction. At the same time, the skewness and kurtosis are always adopted to describe the Non-Gaussian characteristic. Therefore it is necessary to obtain the relationship between the reduction effects of appurtenances on wind pressure and Non-Gaussian characteristic by researching the probability density and the first four orders of moment of the distribution.

As mentioned in Sec. 3.2, the negative peak pressure at partition N1 (the upper leading corner of side face) can be severely affected by arranged appurtenances. The probability density of the fluctuating pressure coefficients at the leading corner for each model is further investigated in this study to clarify the effect of appurtenances on the local wind

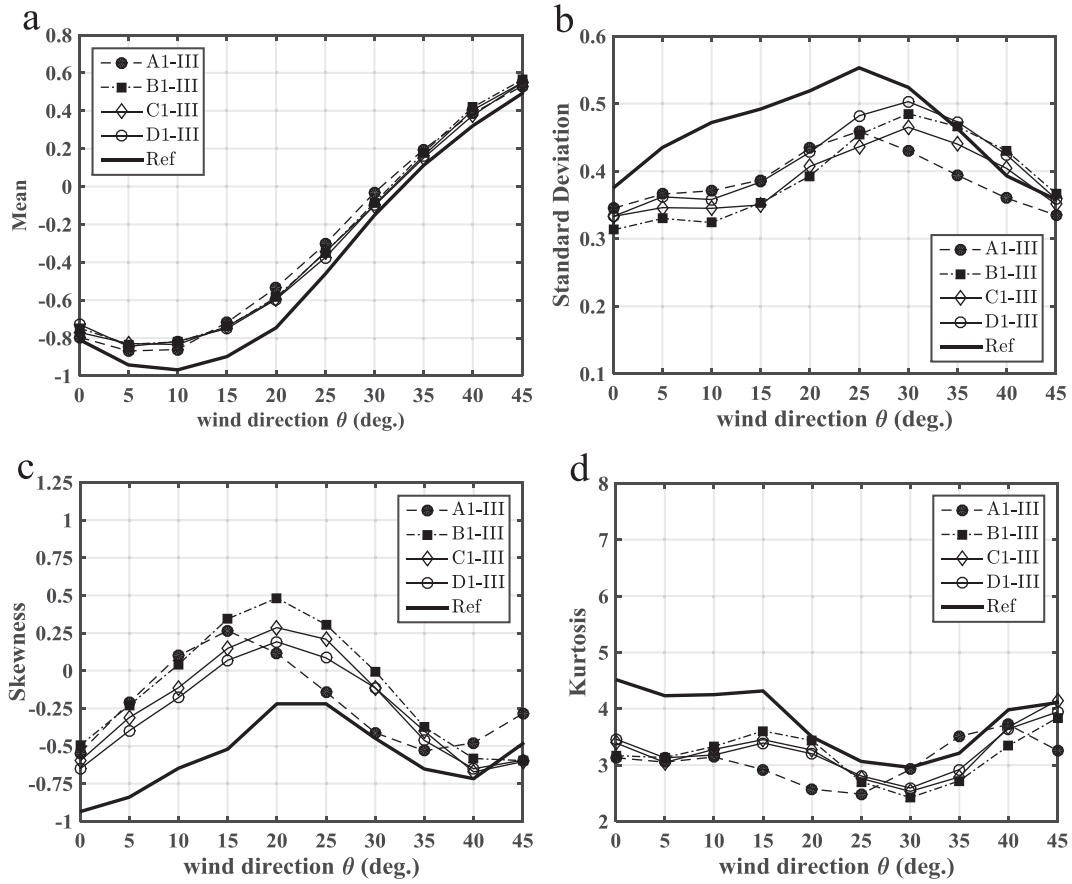


Fig. 13. First four orders of moment of wind pressure coefficients (different  $h_r$ ).

pressure. The probability density distribution of the point with largest negative peak pressure in partition N1 for the models with  $d_r = 12.5\%$  are shown in Fig. 12. The results obtained from most unfavorable wind direction, leading to the largest negative peak pressure in partition N1, is adopted. The result of the model Ref is used for comparison. As shown in Fig. 12, the probability density distribution of the pressure coefficients on reference model tends to have a heavier lower tail than those on models with appurtenances. Such results clearly indicate the reason of a larger negative peak pressures on the reference model. Comparison with the results Fig. 12 (a), the differences between the reference model and the models with appurtenances are relatively indistinctive, as shown in Fig. 12 (b).

To quantify the effects of appurtenances on the probability density, the first four orders of moment of the distribution were examined under different incident wind direction. The results of models with continuous plates can be calculated by Eq. (2) as shown in Fig. 13. It can be seen from Fig. 13 (a) that  $C_{p,mean}(i, \theta)$  for different  $h_r$  almost share the same value when  $d_r = 12.5\%$  for  $\theta \in [0^\circ, 45^\circ]$ , but those values were all smaller than the results of model Ref with maximum difference 15.5%. From Fig. 13 (b), one can also find that the  $C_{p,std}(i, \theta)$  of the models with appurtenances are much lower than that of the reference model. With more close checking on the results, it can be found that the fluctuating pressure coefficients increased with  $h_r$  becoming larger, except for the case  $h_r = 4\%$ . Such results are similar as the extreme pressures shown in Fig. 7 (a). From Fig. 13(c) and (d), the reference model has a skewness close to  $-1$  and a kurtosis over 4.5 when  $\theta = 0^\circ$ . Such results clearly indicate an unsymmetrical non-Gaussian distribution which is also responsible for the large negative peak pressure. It is obvious to note that the presence of the appurtenances can severely affect the third and fourth order of moment by

making the distribution more close to Gaussian, which can further reducing the value of negative peak pressure.

Fig. 14 shows the effect of the horizontal distance ratios  $b_r$  on the probability density distribution of pressure coefficients in partition N1 for  $d_r = 12.5\%$ . Comparing with Fig. 13, one can see that effects of the discontinuous plates are very similar as the continuous plates, but they have relatively weak effects on the distributions especially on mean and STD values. From Fig. 14 (a) and (b), it can be checked that both  $C_{p,mean}(\theta)$  and  $C_{p,std}(\theta)$  become larger when  $b_r$  increase, and the values of model A4-III even almost coincide with those of model Ref. Furthermore, Fig. 14 (c) shows that the probability distributions of pressure coefficients of the models with the larger horizontal gaps show similar skewness as the results of model Ref. The effects on the kurtosis, as shown in Fig. 14(d) are not sensitive to  $b_r$  when  $\theta$  is smaller than  $30^\circ$ . These results explain the reason of relatively weak effects of the discontinuous plates on the negative peak pressure.

#### 4. Conclusion

A series of wind tunnel tests have been conducted to investigate the effect of different façade appurtenances configurations on local wind pressure, including positive and negative peak pressure, the unfavorable wind directions, etc. From this study, the following conclusions can be obtained:

- (1) The largest positive peak pressures of the windward are not significantly affected by different types of appurtenances. But the change of vertical distance ratios ( $h_r$ ) can strongly affect the locations of largest positive peak pressures. Multi-stagnation points can be on the windward wall when  $h_r \in [8\%, 12\%]$ .

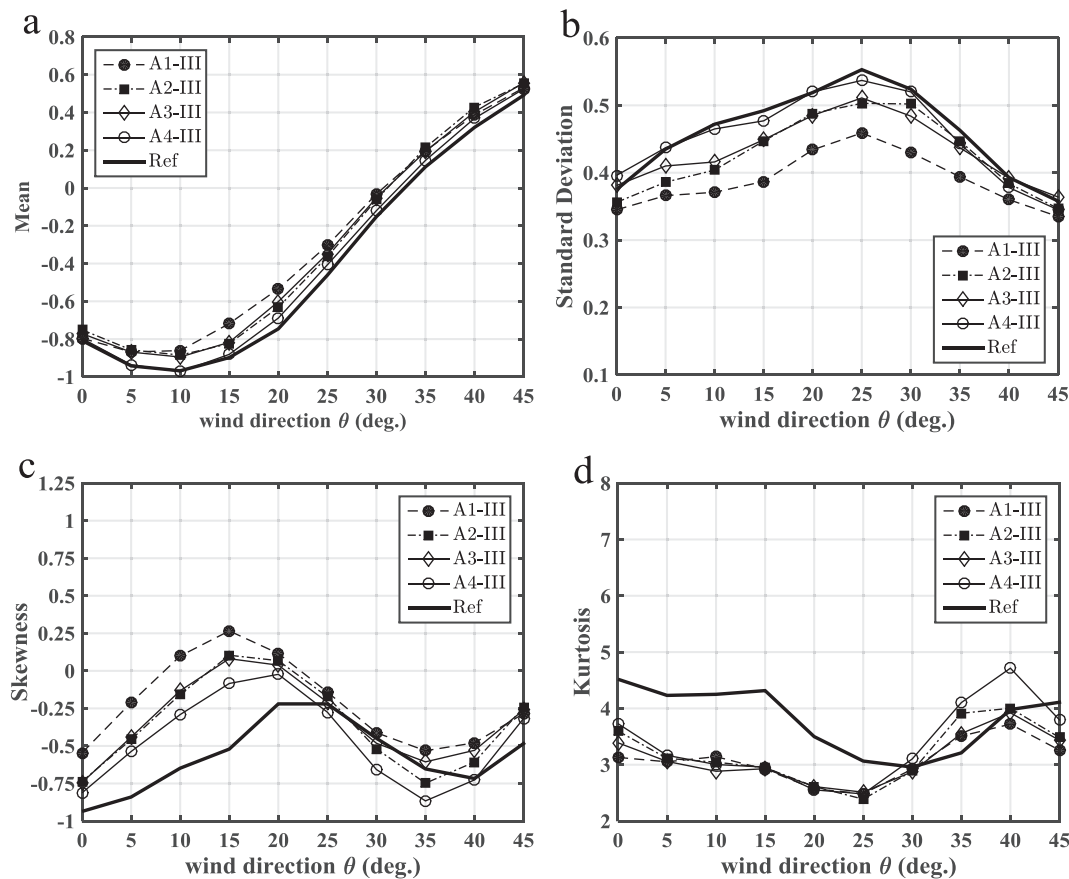


Fig. 14. First four orders of moment of wind pressure coefficients (different  $b_r$ ).

Results also show that when equipped with the appurtenances plates the area with large positive peak pressures ( $C_{p,max}(i) > 1.8$ ) can be greatly reduced by 27% at most from the reference model.

- (2) The unfavorable wind directions of negative peak pressure for both smooth facade model and models with appurtenances are between  $0^\circ$  and  $15^\circ$ , while it is more close to  $0^\circ$  for the models with appurtenances. The largest negative peak pressure for the upper parts of buildings is mostly located at the leading corner for different models. Comparison with the smooth facade model shows that the studied appurtenances configurations have slightly effects on the local peak pressure at the lower leading edge but can remarkably reduce those values at upper leading corner on the side face. The decrement of the largest negative peak pressure coefficients varies from around 22%–42% for different appurtenances arrangements.
- (3) Another effect of the appurtenances is restraining the variations of negative peak pressure on the whole side face. Both the continuous and discontinuous plates can reduce the mean, STD and maximum values of reduction factor, with the effect of discontinuous appurtenances slightly weaker. When  $h_r$  equals to 8% and appurtenances are continuous, the above effect is most significant and the reduction of  $RF_{std}$  even reaches to 60% compared to reference model.
- (4) By investigating the effects of appurtenances on the probability density distribution through the first four order of moment of the probability distribution, it is found that appurtenances can reduce mean and STD of the probability density from the smooth facade model condition, and make the distribution more close to Gaussian distribution. Such changes further explain the reduction

of the negative peak pressure when appurtenances are equipped as studied in this paper.

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